DATA ANALYTICS REPORT

EXPLORATORY DATA ANALYSIS OF SAAS PRODUCT USAGE & CUSTOMER RETENTION

Subscription Behaviour, Product Usage & Customer Retention

TOOLS USED

Python, Pandas, Matplotlib, Seaborn & Plotly

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PROBLEM STATEMENT & OBJECTIVE

Problem Statement:

Modern SaaS businesses operate across diverse customer segments with varying usage and retention behaviors, making it challenging to pinpoint the engagement and operational factors that influence churn and long-term revenue.

Objective:

To explore the SaaS customer usage and retention dataset to uncover patterns, insights, and trends that can help businesses make data-driven decisions around customer engagement, retention, and revenue growth.

EDA WORKFLOW

For this analysis, a structured workflow was followed, involving data collection, understanding, cleaning, exploration, and summarization of insights to allow a clear understanding of the dataset and its trends.

01 Data Collection

- Gathered the Google Play Store dataset from Kaggle

02 Data Understanding & Anomaly Detection

- Looked at data distributions
- Found missing values, outliers, and unusual patterns

03 Data Cleaning & Treatment

- Fixed missing or incorrect values
- Standardized formats for consistency

04 Exploratory Analysis

- Univariate Analysis: e.g Customer Segment, Complaint Type, Contract Type etc
- Bivariate Monthly Logins vs Features Used
- Multivariate Analysis: CSAT vs Contract Type vs Customer Segment

05 Insights & Reporting

- Summarized patterns and trends
- Highlighted key findings for developers and businesses

KEY QUESTIONS EXPLORED

- How do customer demographics and segments influence product usage and retention patterns?
- Which stages of the customer lifecycle contribute most to churn risk?
- Do low-engagement customers represent a larger retention opportunity than highly active users?
- How strongly does product usage depth (logins and feature adoption) influence churn outcomes?
- What role do subscription and pricing structures play in customer commitment and churn?
- How does billing friction, such as payment failures, affect retention?
- How do customer support interactions and resolution quality impact retention and satisfaction?
- To what extent do satisfaction metrics (CSAT and NPS) act as early warning signals for churn?

DATA OVERVIEW

The dataset provides insights into SaaS customer behavior, covering demographics, subscription characteristics, product usage, billing activity & support interactions. It also captures customer engagement signals, satisfaction metrics, and feedback indicators that are relevant for understanding retention and churn dynamics.

Data Source: Kaggle

Dataset Size

10000

Records

32

Features

Purchase Diversity

3

Customer Segments

7

Countries

3

Contract Types

DATA OVERVIEW

Below is a detailed description of the feature set:

Dataset Features	Type	Feature Description
customer_id	Date	Unique identifier assigned to each customer
gender	Categorical	Gender of the customer
age	Numerical	Age of the customer in years
country	Categorical	Country where the customer is located
customer_segment	Categorical	Business-defined customer classification (Individual, SMB, Enterprise)
tenure_months	Numerical (Discrete)	Number of months the customer has been subscribed to the service
signup_channel	Categorical	Channel through which the customer signed up (Organic, Ads, Referral)
contract_type	Categorical	Type of subscription contract (Monthly, Quarterly, Annual)
monthly_logins	Numerical (Discrete)	Total number of platform logins by the customer in a month
weekly_active_days	Numerical (Discrete)	Average number of days per week the customer actively used the platform
avg_session_time	Numerical (Continuous)	Average duration of a customer session on the platform
features_used	Numerical (Discrete)	Number of distinct product features used by the customer
usage_growth_rate	Numerical (Continuous)	Percentage change in usage over a recent time period
last_login_days_ago	Numerical (Discrete)	Number of days since the customer last logged into the platform
monthly_fee	Numerical (Discrete)	Monthly subscription fee charged to the customer
total_revenue	Numerical (Discrete)	Total revenue generated from the customer over their lifetime

DATA OVERVIEW

Continued..

Dataset Features	Type	Feature Description
city	Categorical	City where the customer resides
payment_method	Categorical	Payment method used by the customer
payment_failures	Numerical (Discrete)	Number of failed payment attempts recorded
discount_applied	Categorical	Indicates whether a discount was applied to the subscription
price_increase_last_3m	Categorical	Indicates whether the customer experienced a price increase in the last three months
support_tickets	Numerical (Discrete)	Number of customer support tickets raised
avg_resolution_time	Numerical (Continuous)	Average time taken to resolve customer support tickets
complaint_type	Categorical	Category of complaint raised by the customer
csat_score	Numerical (Continuous)	Customer Satisfaction Score
escalations	Numerical (Discrete)	Number of support cases escalated beyond first-level support
email_open_rate	Numerical (Continuous)	Percentage of marketing or communication emails opened
marketing_click_rate	Numerical (Continuous)	Percentage of marketing emails where links were clicked
nps_score	Numerical (Discrete)	Net Promoter Score provided by the customer
survey_response	Categorical	Indicates whether the customer responded to surveys
referral_count	Numerical (Discrete)	Number of new customers referred by the existing customer
churn	Numerical (Binary)	Indicates whether the customer discontinued the service (0 = No, 1 = Yes)

DATA QUALITY CHALLENGES & ANOMALIES

Few inconsistencies were found in the dataset, which could have affected the analysis if left unaddressed.

DATA ANOMALIES

- nps_score is stored as an int64. While numeric, it functions as an ordinal / categorical business metric (e.g., Detractors, Passives, Promoters). Depending on analytical objectives, it may require grouping or categorical transformation rather than being treated as a continuous variable.
- Several categorical attributes are stored as object (string) types, including: gender, country, city, customer_segment, signup_channel, contract_type, payment_method, complaint_type, discount_applied, price_increase_last_3m, and survey_response. Converting these fields to the category data type would improve analysis without altering semantic meaning.
- complaint_type contains 2,045 missing values (20.45%)
- No duplicate or conflicting columns were observed in the final dataset.

DATA CLEANING & TREATMENT

Inconsistent and missing values were addressed, and key features were cleaned and standardized for analysis.

DATA CLEANING SUMMARY

- Several categorical features, including gender, country, city, customer_segment, signup_channel, contract_type, payment_method, complaint_type, discount_applied, price_increase_last_3m, and survey_response, were standardized through data type conversion to ensure consistency and improved analytical performance.
- The **complaint_type** column contained missing values representing customers who did not raise any complaints. Rather than imputing these missing values with the mode (which would introduce bias), missing entries were replaced with a meaningful category such as "No Complaint".

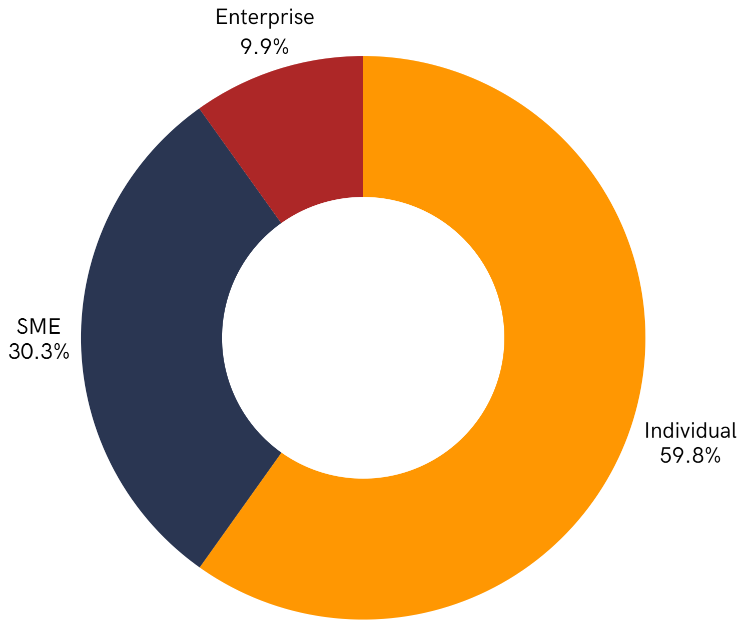
INSIGHTS

Customer Profile & Account Overview

Customer base is heavily skewed toward Individuals, while Enterprise adoption remains limited

59.8%

Of the customers are individuals



Key observations

- Individual customers make up the largest share of the user base, far exceeding both SME and Enterprise segments.
- Enterprise customers represent a very small portion of total users, indicating low penetration compared to other segments.

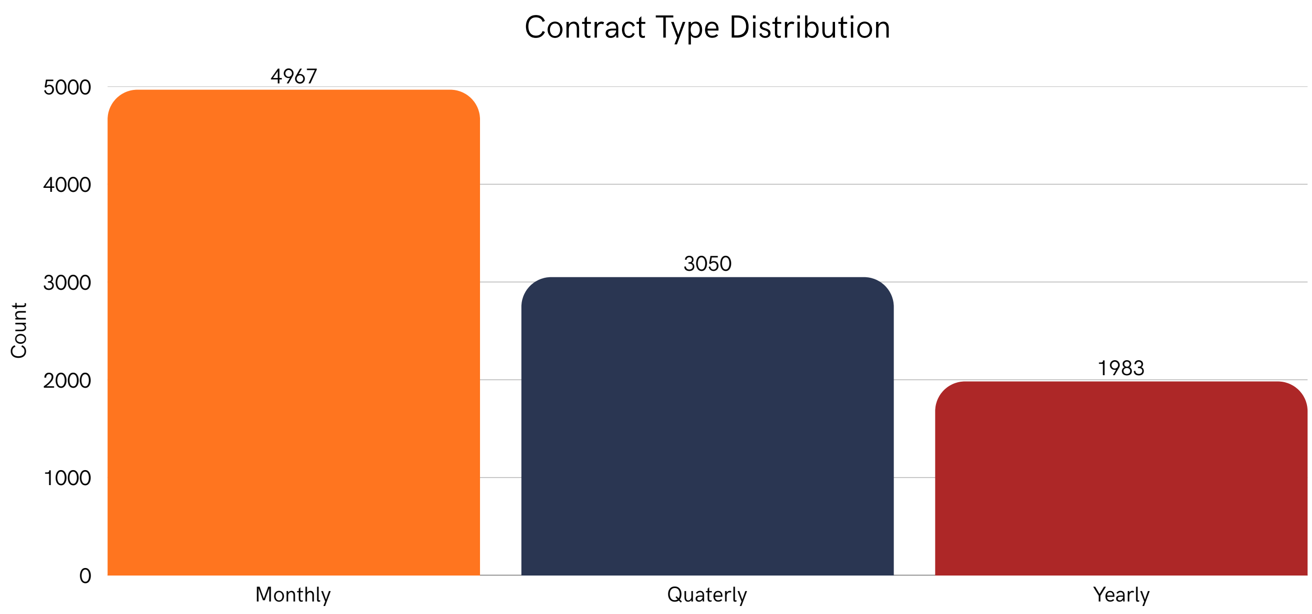
Business Insights

- Strong Individual adoption suggests the product is well-positioned for self-serve usage, but long-term revenue growth may be constrained without deeper SME and Enterprise expansion.
- The limited Enterprise presence highlights a need to reassess enterprise-specific offerings—such as pricing, features, or sales strategy—to unlock higher-value customer growth and improve revenue stability.

Short-term contracts dominate customer subscriptions, limiting long-term revenue stability

49.67%

Of the contracts are short-term (monthly)



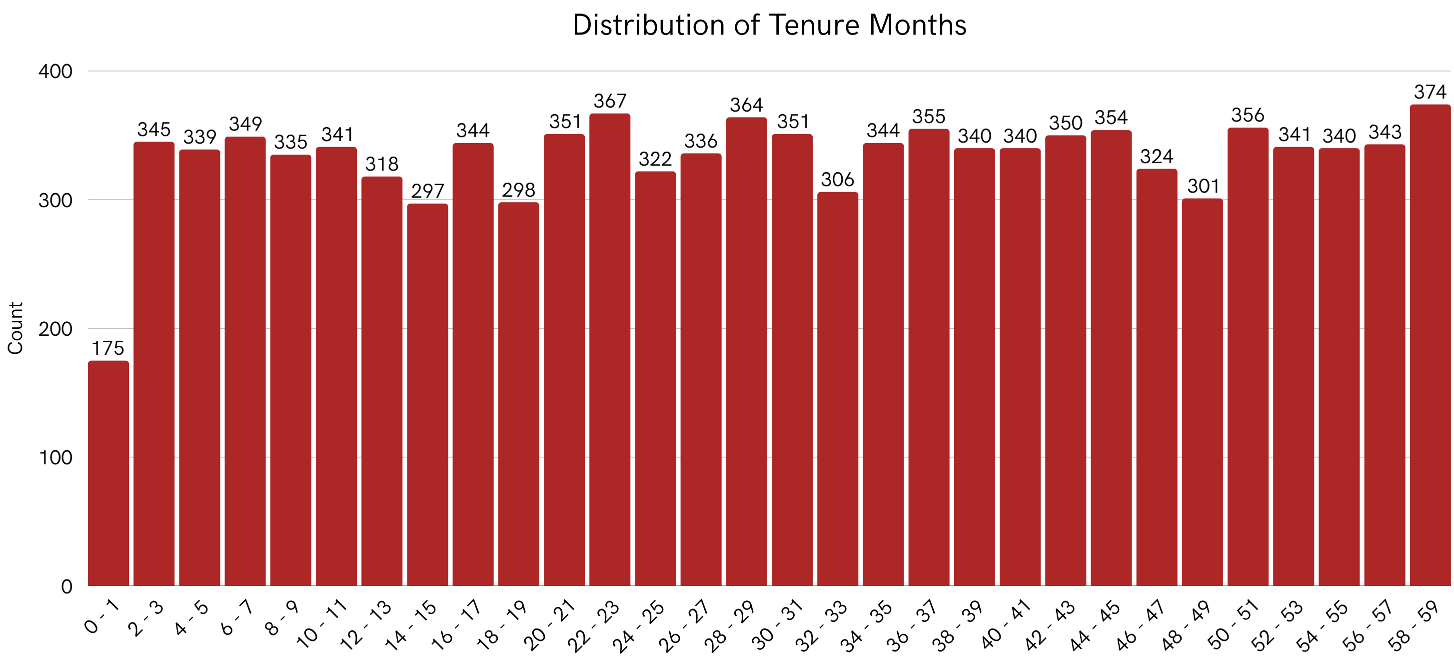
Key observations

- Monthly contracts are the most common subscription type, significantly outnumbering both Quarterly and Yearly plans.
- Yearly contracts have the lowest adoption, indicating limited commitment to long-term subscriptions.

Business Insights

- Heavy reliance on monthly contracts increases churn exposure and makes revenue less predictable over time.
- Encouraging upgrades to Quarterly and Yearly plans through pricing incentives or value bundling could improve retention and stabilize long-term revenue.

Customer tenure is broadly distributed, indicating steady retention beyond early lifecycle stages



Key observations

- Customers are spread relatively evenly across tenure months, with no sharp drop-off in the early subscription period.
- A substantial portion of users remain active well beyond the first year, extending toward long-term tenures.

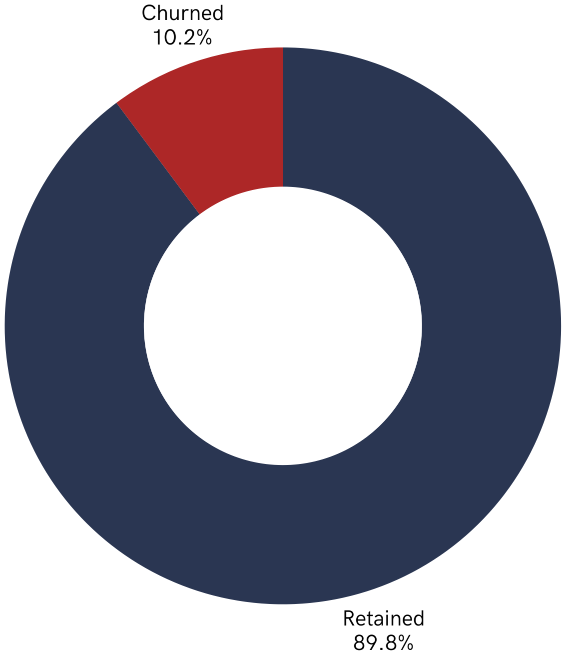
Business Insights

- The absence of early tenure drop-offs suggests effective onboarding and early product adoption.
- Since customers persist across long tenures, improving engagement and expansion strategies for mid- to long-term users could further strengthen lifetime value and reduce long-term churn risk.

Overall churn remains low, indicating strong baseline customer retention

89.8%

Customers Retention Rate



Key observations

- The vast majority of customers are retained, with churned users representing only a small fraction of the total customer base.
- There is a clear imbalance between retained and churned customers, showing that most users continue their subscriptions.

Business Insights

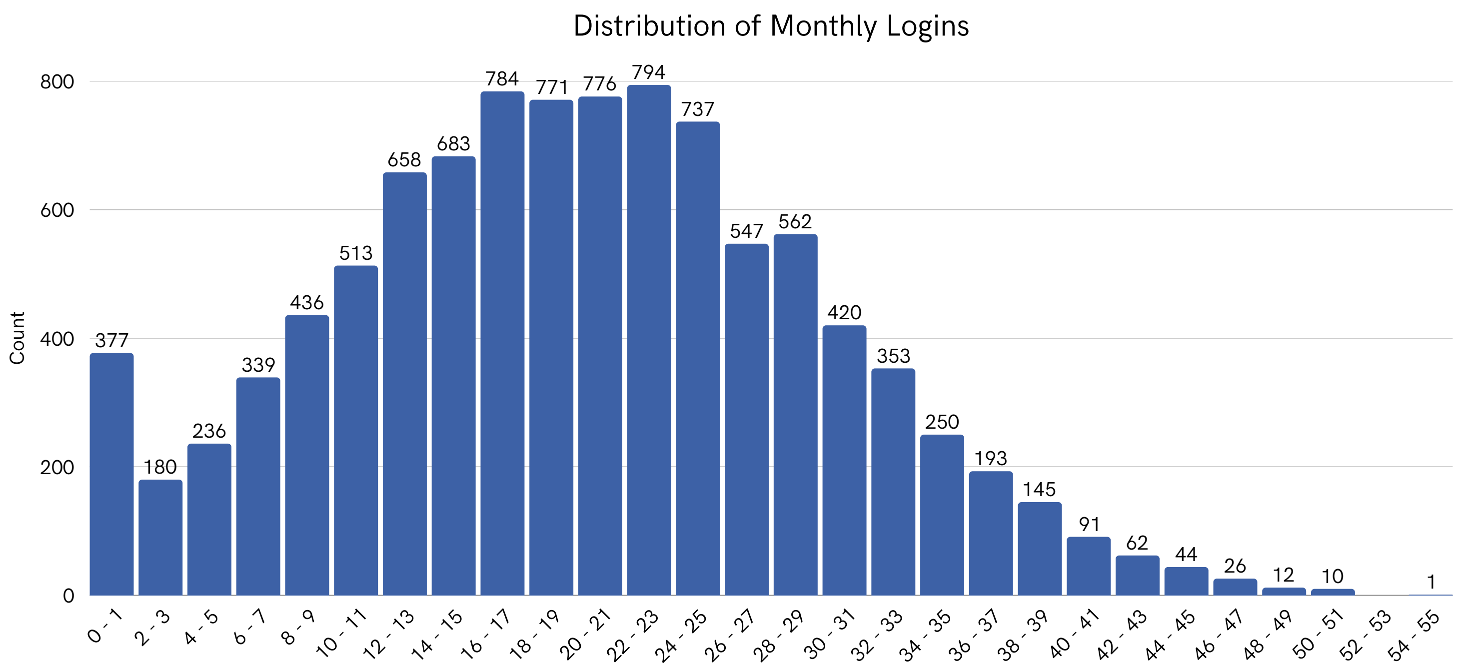
- Low overall churn suggests the product delivers sufficient ongoing value for most customers.
- Despite strong retention at an aggregate level, targeted analysis is needed to identify high-risk subgroups, as small churn segments can still have an outsized impact on long-term revenue and growth.

Product Adoption & Usage Health

Most customers show moderate login activity, with a small group of highly engaged power users

20

Typical Monthly Logins (Median)



Key observations

- Monthly logins are concentrated in the mid-range, indicating consistent but not daily usage for most customers.
- A long right tail exists, showing a smaller segment of users logging in very frequently compared to the majority.

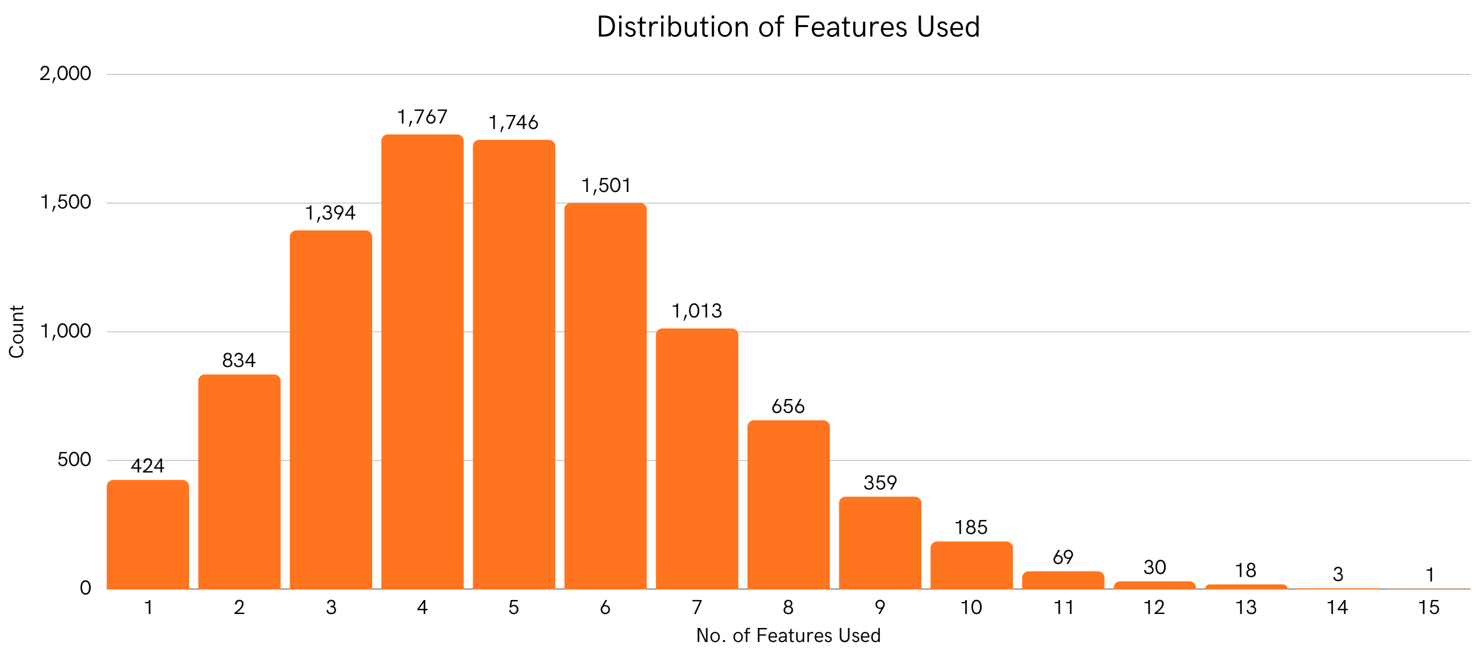
Business Insights

- The concentration of users in the moderate activity range suggests healthy baseline engagement but room to deepen habitual usage.
- Highly active users likely represent power users or high-value accounts, making them strong candidates for upsell, advocacy, and referral-driven growth strategies.

Most customers engage with a limited set of core features, while deep feature adoption remains rare

3 - 6

Typical Number of Features Used



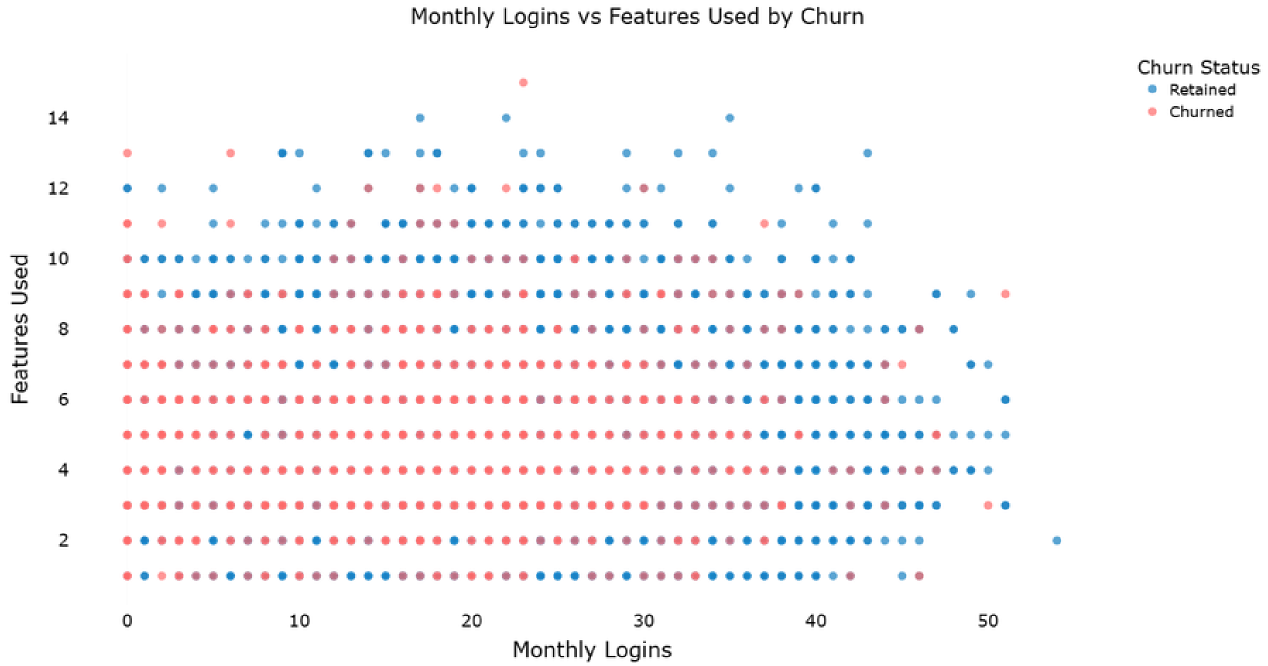
Key observations

- Feature usage is heavily concentrated around a small number of features, with most customers using roughly 3–6 features.
- Only a small minority of users adopt a wide range of features, creating a long tail of advanced usage.

Business Insights

- Concentrated usage around core features suggests these drive primary product value, but many customers may not be fully realizing the platform’s capabilities.
- Improving feature discovery, onboarding, and in-product guidance could expand feature adoption and increase customer stickiness, helping reduce long-term churn risk.

Low engagement intensity strongly aligns with churn, while retained users show broader and more frequent product usage



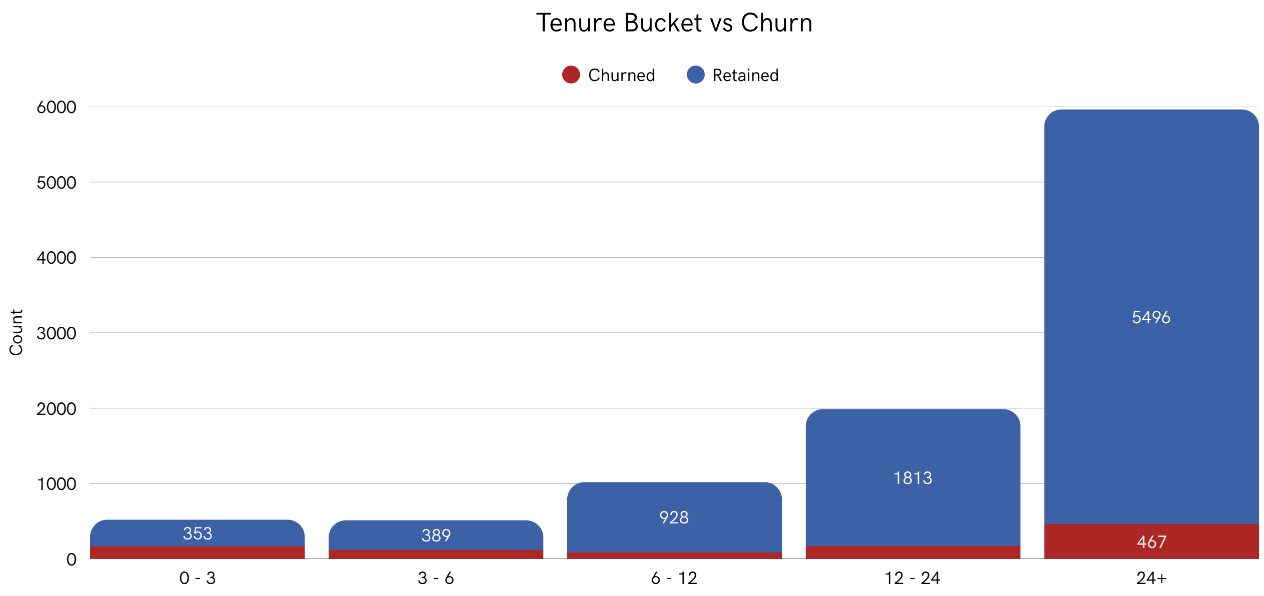
Key observations

- Churned customers are heavily concentrated at lower monthly logins and fewer features used.
- Retained customers are more prevalent at higher login frequencies and higher feature adoption levels.

Business Insights

- Customers who fail to build regular usage habits and explore multiple features are significantly more likely to churn.
- Driving early engagement through activation nudges, feature discovery, and usage-based onboarding can materially reduce churn risk by moving users into higher-engagement zones.

Churn risk is front-loaded in early tenure, while long-tenured customers show strong retention stability



Key observations

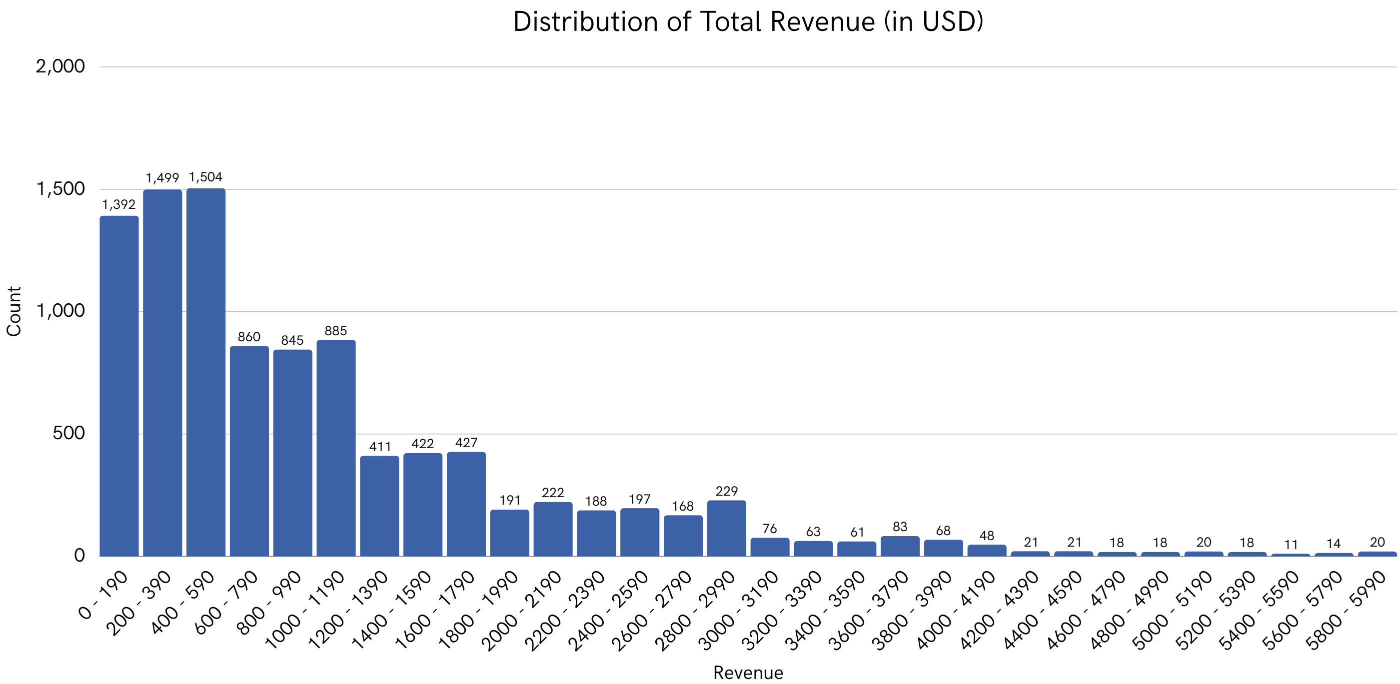
- Churn is proportionally higher in the early tenure buckets (0-3 and 3-6 months) compared to later stages.
- Customers with longer tenure, especially 24+ months, are overwhelmingly retained with minimal churn presence.

Business Insights

- The first few months represent the most critical window for preventing churn, making onboarding and early activation essential.
- Once customers cross long-term tenure thresholds, retention stabilizes, indicating that efforts to push users past early lifecycle stages can significantly improve lifetime value.

Revenue & Pricing Sensitivity

Revenue is concentrated among a small group of high-value customers, while most generate modest lifetime value



Key observations

- The total revenue distribution is heavily right-skewed, with most customers contributing lower lifetime revenue.
- A long tail of customers generates significantly higher revenue, indicating the presence of a small high-value segment.

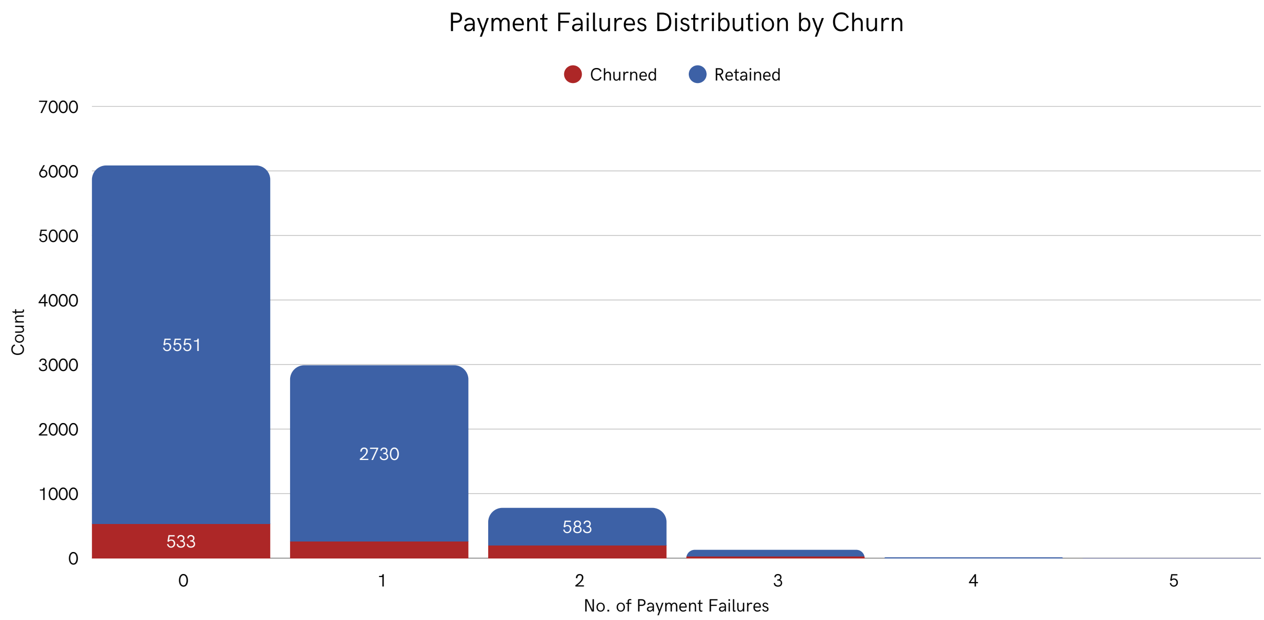
Business Insights

- A minority of customers likely drive a disproportionate share of total revenue, making their retention critical to overall business performance.
- Identifying the behaviors and characteristics of these high-value customers can inform targeted retention, upsell, and expansion strategies to maximize lifetime value.

Early payment failures show a visible association with increased churn

2 - 3

Typical Number of Payment Failures (Median)



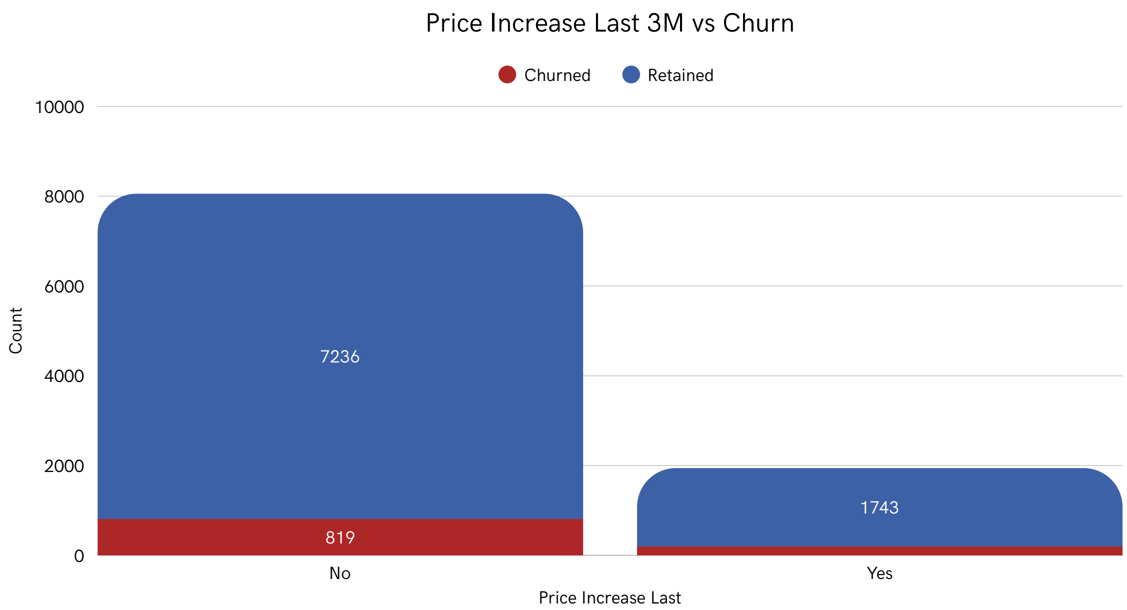
Key observations

- Most retained customers have zero or only one payment failure.
- Churned customers are disproportionately represented as the number of payment failures increases.

Business Insights

- Payment friction appears to be a key driver of involuntary churn.
- Proactive interventions such as payment retry logic, early alerts, and alternative payment options could significantly reduce avoidable customer losses.

Price Increases Show Minimal Impact on Customer Churn



Key observations

- Churn levels among customers who experienced a price increase remain relatively low compared to retained customers.
- The majority of churn comes from customers who did not face a recent price increase.

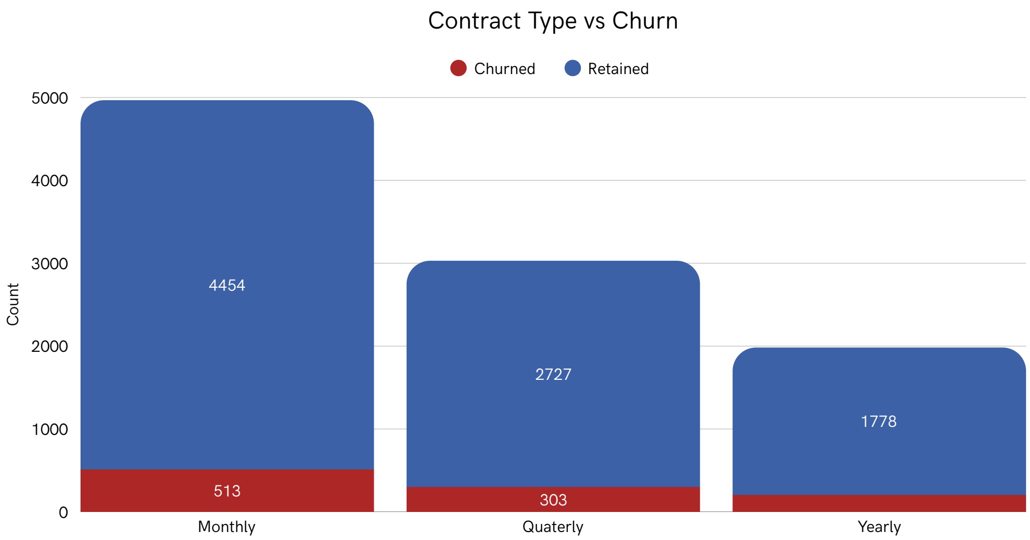
Business Insights

- Recent price adjustments do not appear to be a major churn trigger, suggesting strong perceived product value.
- This indicates potential room for strategic pricing optimization without significantly harming retention.

Short-Term Contracts Drive Higher Churn Risk Compared to Long-Term Plans

303

Typical Number of Churns per Contract (Median)



Key observations

- Monthly contracts show the highest churn volume compared to Quarterly and Yearly plans.
- Yearly contracts exhibit the lowest churn levels, with significantly stronger retention.

Business Insights

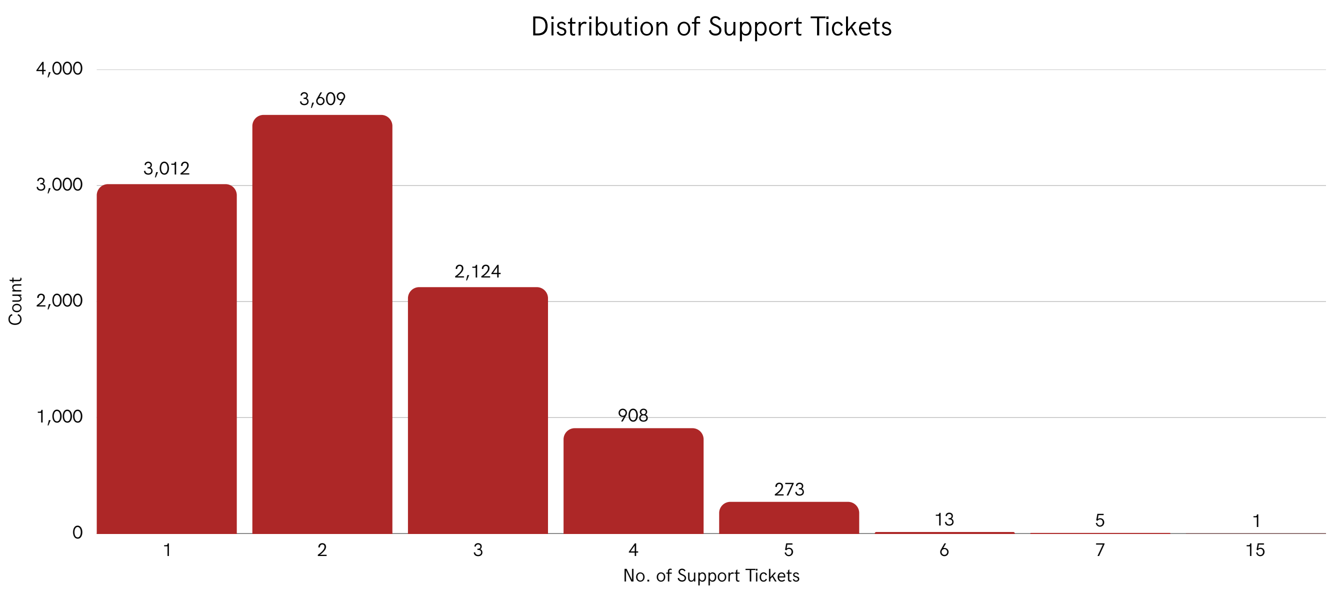
- Customers on short-term (Monthly) plans are more prone to churn, indicating lower commitment levels.
- Incentivizing upgrades to longer-term contracts (Quarterly/Yearly) could materially improve retention and revenue stability.

Customer Support & Experience Quality

Customer Support Demand Is Concentrated Among Low-Ticket Users

66.21%

Of the Customers raise 0 - 1 Support Tickets



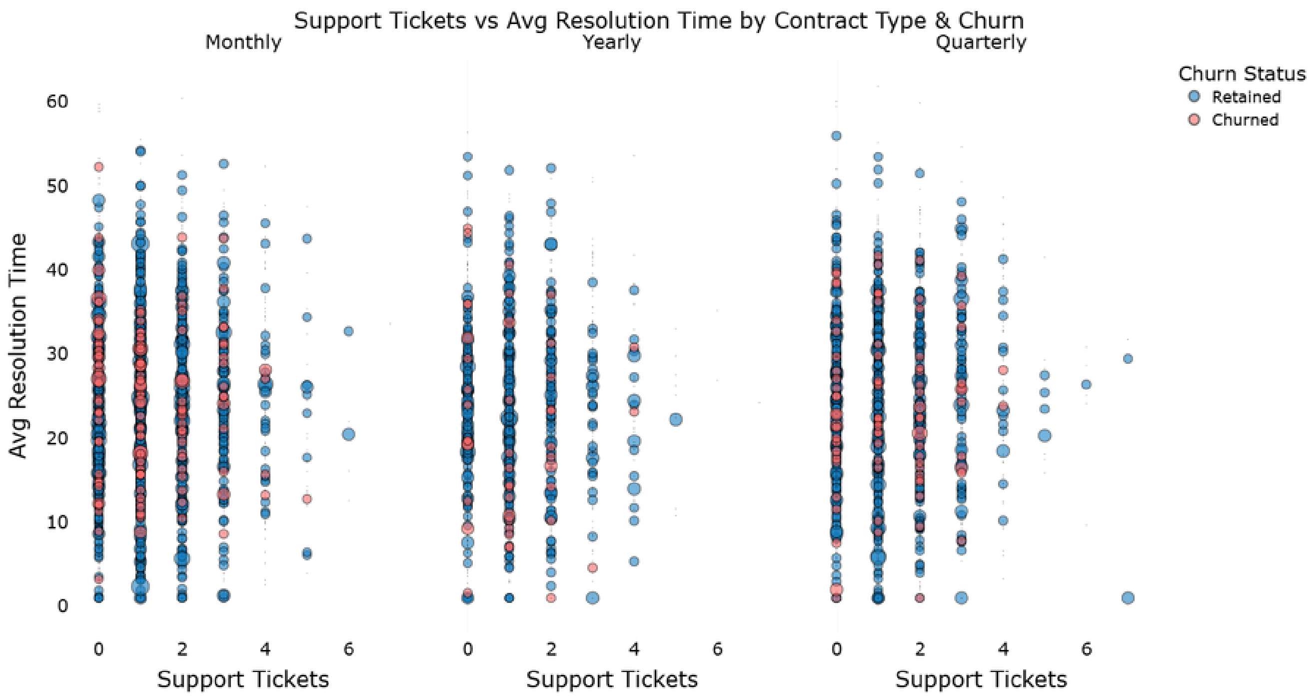
Key observations

- The majority of customers raise 0-1 support tickets, with counts declining sharply beyond 2 tickets.
- Very few customers generate high ticket volumes (4+), indicating a small segment of heavy support users.

Business Insights

- Most customers experience minimal friction, suggesting overall product stability.
- A small high-ticket segment may represent at-risk or high-maintenance customers, requiring proactive support or product fixes to prevent churn.

Longer Resolution Times combined with Frequent Support Tickets significantly increase Churn Risk, especially among Monthly Contracts



Key observations

- Churn correlates with higher support ticket counts and longer resolution times, with Monthly customers showing the highest sensitivity, Yearly customers the lowest, and Quarterly in between.
- Yearly customers retain better than Monthly or Quarterly even at similar support volumes, highlighting contract length’s buffering effect on churn.

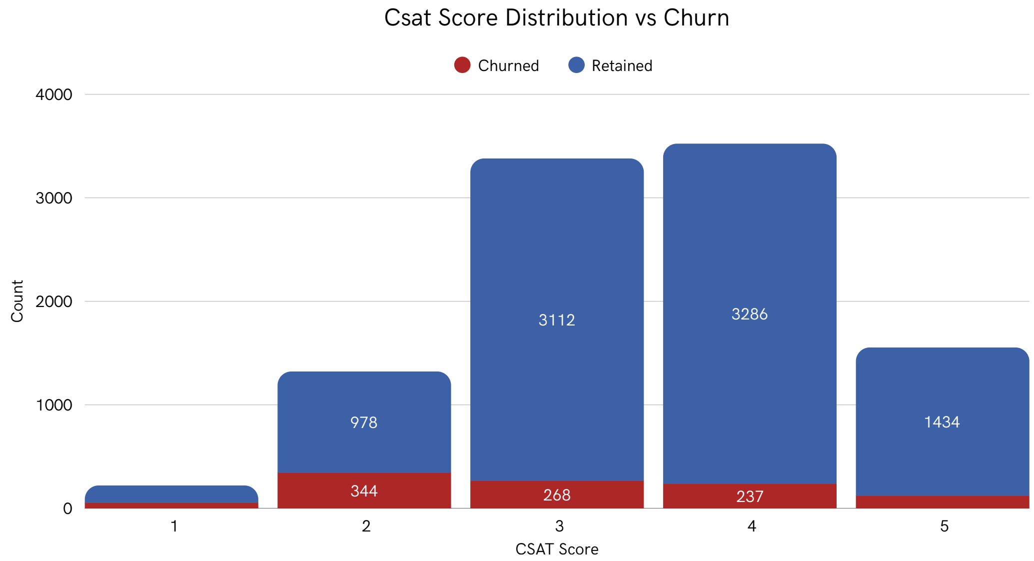
Business Insights

- Support friction (both ticket volume & slow resolution) is a clear churn driver, especially for Monthly plans with low commitment.
- Faster resolution and proactive support (SLAs, escalation, early intervention) can reduce churn significantly, potentially without changing pricing or features.

Strong retention signal here. Very clean.

69.03%

Of the Customers have a CSAT score between 3 - 4



Key observations

- Customers with low CSAT (1-2) have much higher churn, while the majority of retained customers score 3–5, with very few churning at 5.
- Churn steadily declines as satisfaction increases, showing strong loyalty at high CSAT scores.

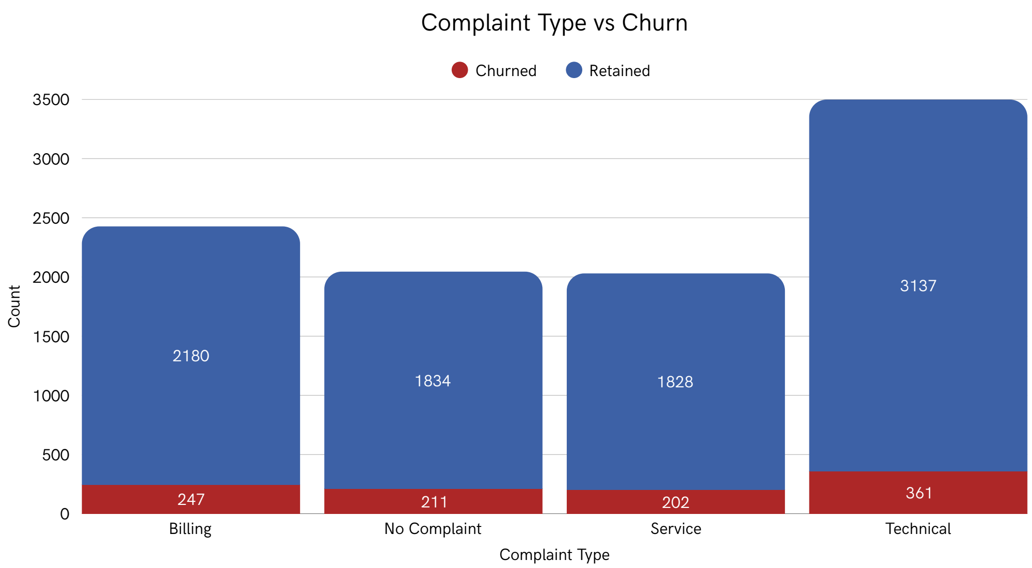
Business Insights

- Customer satisfaction is a leading churn indicator; even moderate dissatisfaction (2-3) increases retention risk.
- Raising CSAT by one point and using low scores (≤2) as early-warning triggers can meaningfully reduce churn through proactive outreach.

Technical Complaints Drive the Highest Churn Risk Compared to Other Complaint Types

43.97%

Of the Complaints are Technical issues



Key observations

- Technical issues drive the highest churn, followed by billing and service complaints; even customers with no complaints churn, though less frequently.
- Technical problems are the most critical friction point affecting retention.

Business Insights

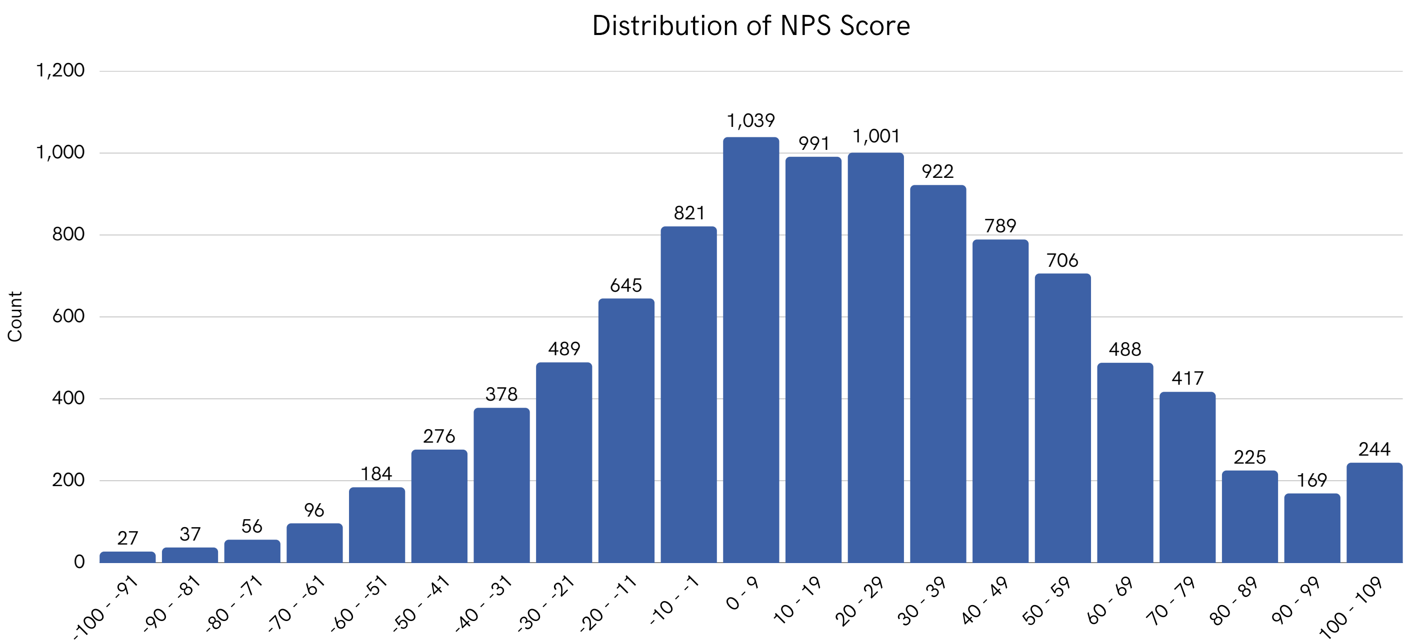
- Prioritizing technical issue resolution through product stability, bug fixes, and proactive monitoring can significantly reduce churn.
- While billing/service complaints matter, differentiated support strategies and holistic retention efforts are needed for “no complaint” churners.

Engagement, Loyalty & Advocacy

Customer Sentiment Skews Neutral-to-Positive, but a Meaningful Detractor Base Signals Churn Risk

19.0

Typical NPS Score (Median)



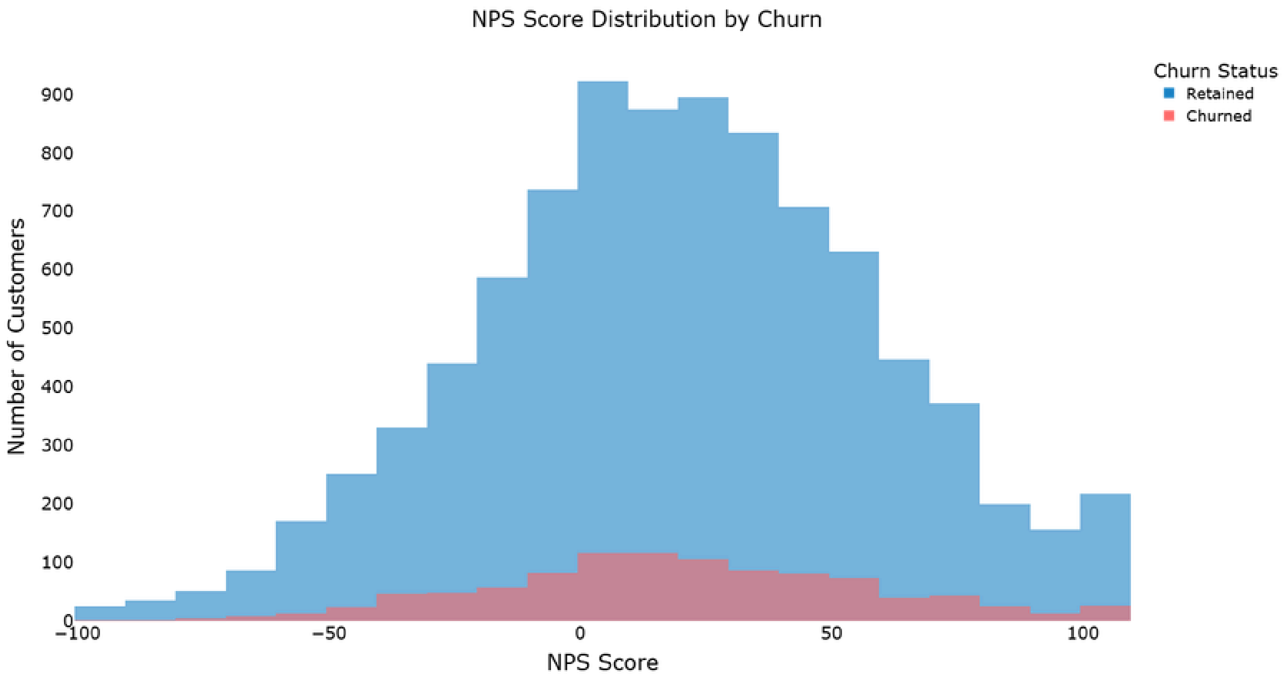
Key observations

- The NPS distribution is concentrated in the mid-range (0-40), indicating a large passive customer base, with fewer extreme high promoter scores (80-100), forming a broad bell-shaped curve rather than a strong right skew.
- There is a clear cluster of negative scores (below 0), showing a meaningful detractor segment alongside the dominant passive group.

Business Insights

- The large passive base is a major conversion opportunity, while the detractor segment signals potential churn risk, especially if linked to low usage, support issues, or pricing changes.
- The limited dominance of strong promoters suggests the product is satisfactory but not highly differentiated, meaning improvements in onboarding, feature adoption, and support experience could shift the curve right and increase retention and referrals.

Low NPS Scores Strongly Correlate with Churn, While Promoters Show Higher Retention Stability



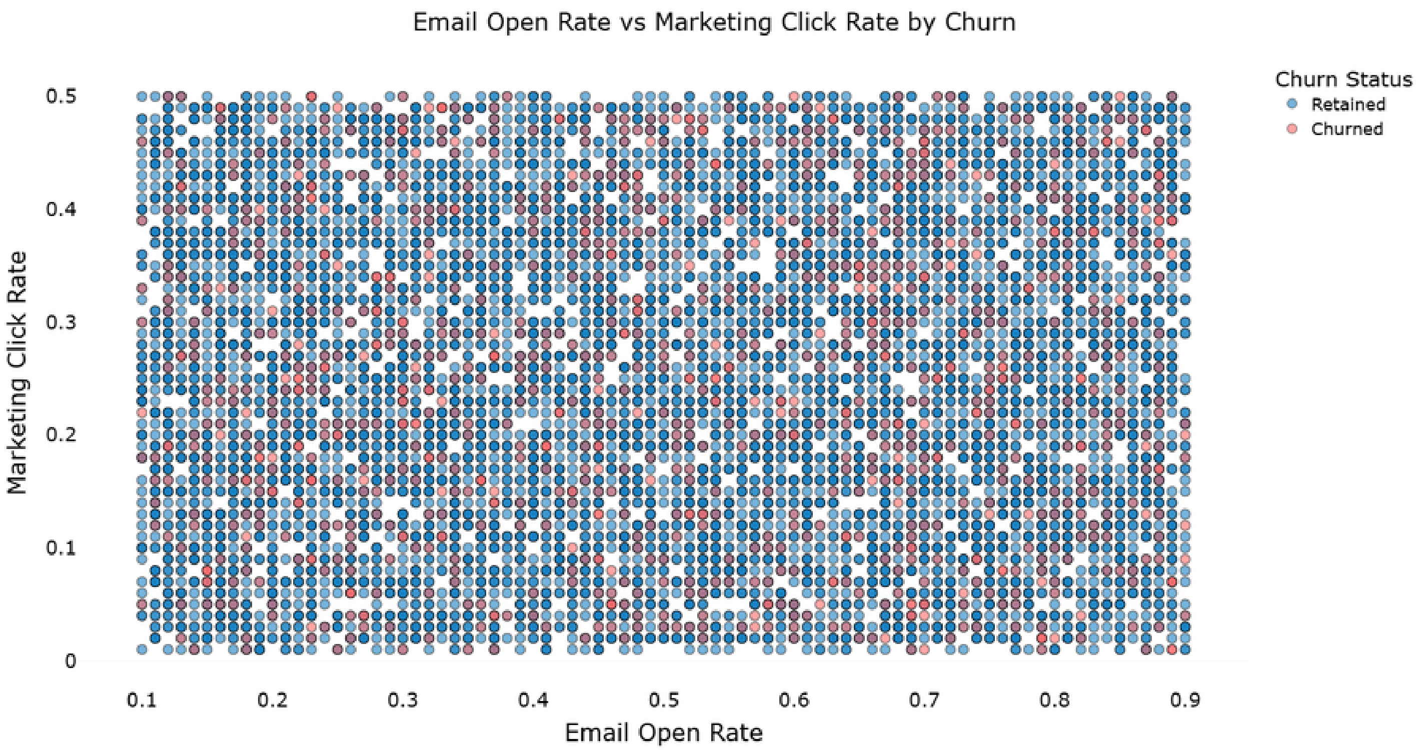
Key observations

- Customers who churn are mostly in the negative and low NPS ranges, while retained customers dominate mid to high NPS scores (20–60+), with very high scores (80–100) almost entirely retained.
- The mid-range (0–30) shows overlap between churned and retained customers, highlighting a significant passive segment present in both groups.

Business Insights

- NPS is a strong leading indicator of churn; detractors (<0) need proactive intervention, and passives (0-30) are a key conversion opportunity to reduce churn.
- Promoters are high-retention assets and referral drivers, and embedding NPS into churn models can enhance early-risk detection and retention strategies.

Marketing Engagement Alone Does Not Clearly Differentiate Churned vs Retained Customers



Key observations

- Retained and churned customers overlap heavily in email open rates (≈10%–90%) and click rates (≈0%–50%), with no clear low-engagement clustering among churned users.
- High engagement does not guarantee retention, as churn occurs even among users with strong open and click rates.

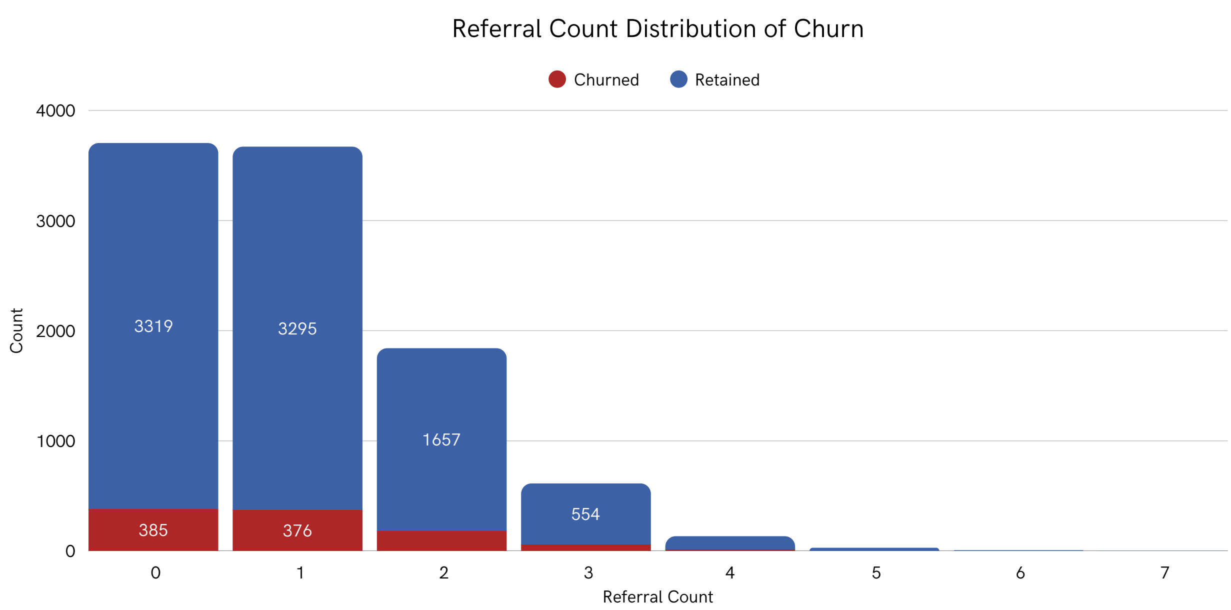
Business Insights

- Email engagement alone is a weak predictor of churn; retention is more influenced by product usage, satisfaction (NPS/CSAT), billing, or support experience.
- Marketing metrics should support broader churn models, and churn among highly engaged users may signal messaging misalignment, feature gaps, or pricing sensitivity.

Higher Referral Activity Strongly Correlates with Customer Retention and Lower Churn Risk

73.75%

Of the Referrals are between 0 - 1



Key observations

- Customers with 0-1 referrals account for the vast majority of churn, while churn drops sharply as referral count increases.
- Customers with 3+ referrals are overwhelmingly retained, indicating a strong inverse relationship between referral activity and churn risk.

Business Insights

- Referral behavior is a powerful proxy for engagement and loyalty, customers who actively refer are significantly less likely to churn, suggesting advocacy reflects deep product value realization.
- Driving referral participation early in the lifecycle can serve as both a growth and retention lever, turning passive users into long-term, high-LTV advocates while helping identify low-referral customers as churn-risk segments.

BUSINESS / DEVELOPER TAKEAWAYS

- **Self-Serve Success, Enterprise Gap**: Strong Individual adoption confirms effective self-serve positioning, but limited SME and Enterprise presence signals a ceiling on revenue growth without segment-specific pricing, features, and sales strategies.
- **Early Lifecycle Experience Determines Retention**: Young adults dominate the user base, making them the primary target for products, messaging & promotions, while simplified UX and trust signals can help retain older users.
- **Engagement Depth Is the Primary Churn Buffer:** Frequent logins and broader feature usage strongly align with retention, reinforcing engagement depth as the core driver of long-term customer stability.
- **Core Features Anchor Value, Expansion Drives Stickiness:** Most customers rely on a small set of core features, but deeper feature adoption is necessary to increase stickiness and reduce long-term churn risk.
- **Monthly Plans Increase Revenue Volatility:** Heavy reliance on monthly contracts exposes the business to higher churn and unstable revenue, while longer-term plans reflect stronger commitment and predictability.

BUSINESS / DEVELOPER TAKEAWAYS

- **High-Value Customers Carry Outsized Impact:** A small segment of customers contributes a disproportionate share of total revenue, making their retention, satisfaction, and expansion strategically critical.
- **Operational Friction Triggers Involuntary Churn:** Payment failures and unresolved support issues are more prevalent among churned customers, highlighting billing reliability and support efficiency as essential safeguards.
- **Pricing Power Exists When Value Is Clear:** Recent price increases do not materially elevate churn, suggesting customers perceive strong value and allowing room for cautious, value-led pricing optimization.
- **Satisfaction Scores Predict Churn Before It Happens:** CSAT and NPS function as leading churn indicators, with detractors requiring immediate intervention and passives representing the largest retention opportunity.
- **Advocacy Signals Deep Loyalty and Retention:** Customers who actively refer others are significantly less likely to churn, making referral behaviour a powerful proxy for long-term value and loyalty.

CHALLENGES & OPPORTUNITIES

Limitations

- The dataset is survey-based and may not fully reflect actual or current Amazon consumer behavior trends.
- Respondents represent a limited subset of Amazon users, potentially introducing bias toward more active or engaged shoppers.
- Timestamped responses provide only a snapshot in time, limiting long-term trend analysis.

Future Work / Opportunities:

- Perform sentiment analysis on customer reviews to understand consumer perception and satisfaction.
- Conduct time-series analysis on purchase and browsing patterns to study how consumer engagement evolves over time.
- Compare frequent buyers with infrequent buyers to identify factors driving purchase behavior.
- Analyze the impact of product categories, personalized recommendations and review metrics on shopping decisions.

CONCLUSION

This analysis of Amazon consumer behavior provided a comprehensive understanding of purchasing patterns, browsing habits, and engagement with platform features. Key insights and trends were identified, demonstrating how data-driven analysis can inform strategic decisions for sellers, marketers and platform managers. Future analysis could include sentiment analysis of customer reviews, time-series tracking of purchase and browsing patterns and comparisons between frequent and infrequent buyers to uncover deeper insights.

This analysis was conducted using Python, Pandas, Matplotlib, Seaborn and Plotly.

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